

The Impact of Parental Educational Attainment on Child Labor

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Abstract

Child labor remains a significant global challenge that affects children's education, health, and overall well-being. This paper investigates the relationship between parental educational levels and child labor in five African nations using UNICEF's Multiple Indicator Cluster Surveys (MICS6) data. With 160 million children worldwide engaged in child labor and one third of sub-Saharan African child laborers involved in hazardous work, understanding household-level determinants is crucial for policy development (ILO, 2024). The theoretical framework models parental decision making as balancing household welfare with children's current and future welfare, where education influences these preferences. I explore the influence of parents' educational attainment on the number of hours their children work weekly on economic and none-economic activities. The primary findings reveal a significant negative relationship between the level of education of parents and the number of weekly hours their children work. For fathers with tertiary education, their children working 11% fewer hours than children of uneducated fathers and mothers with elementary education, 8% fewer hours. The study develops a novel framework explaining how parental education affects time allocation decisions for children. Findings suggest that promoting adult education—particularly for mothers—could serve as an indirect but effective strategy to combat child labor while addressing its root causes within household decision-making structures.

KEYWORDS: Child labor, Parental Education, School Attendance, Malawi, Nigeria, Ghana, Gambia, Zimbabwe.

JEL codes: J13, J22, I24, I32

Introduction

Child labor continues to be a widespread challenge, particularly in Sub-Saharan Africa, depriving children of their childhood, education, and opportunities for future development. The International Labor Organization estimates that 160 million children worldwide are victims of child labor where 79 million engage in hazardous work (ILO, 2024). In sub-Saharan Africa, more than one-third of children in labor are involved in unsafe work, particularly in the artisanal mining and agricultural sectors (Mbah-Amanze, 2024). The long-run effects of children's early entry into the labor market harm the child's well-being and the economic outlook of the country.

The importance of this research lies in understanding the household-level determinants of child labor, particularly the role of parental education in shaping a child's labor and schooling outcomes. Research shows that household characteristics like income can also affect the number of hours a child works. Financial hardship contributes significantly by limiting parents' ability to provide for their children, forcing them to work for pay. The technical demands of modern labor markets only reinforce the arguments that more years of schooling lead to higher pay, which in turn reduces the number of hours the child works on economic activities.

Educated parents are more likely to be aware of the long-term benefits of schooling and the legal implications of child labor, which may reduce their children's likelihood of working. Conversely, low parental education may lead to a lack of awareness and negligence, increasing the probability of children working long hours weekly. Using the UNICEF's Multiple Indicator Cluster Surveys (MICS), this paper explores how parental education protects against child labor. Understanding this relationship is crucial for policymakers, as interventions to increase adult education and parental awareness could serve as indirect strategies to reduce child labor. This paper uses a multiple linear regression model with MICS household survey data for 5 African countries over 5 years.

The microeconomic framework of this study conceptualizes child labor as the outcome of parental decision-making regarding time allocation. Parents maximize a utility function that balances current household welfare with their perception of the child's welfare, expressed as $U_p = \alpha \cdot W_c + (1 - \alpha) \cdot W_h$, where α increases with parental education. The child's welfare (W_c) comprises both current utility (which decreases with labor hours) and expected future welfare (which increases with schooling). Parental education (E_p) influences two key parameters: the weight placed on the child's welfare relative to household welfare ($\frac{\partial \alpha}{\partial E_p} > 0$) and the emphasis on long-term versus short-term outcomes ($\frac{\partial \beta}{\partial E_p} < 0$). This creates a mechanism where more educated parents recognize the future returns to education and consequently reduce children's labor hours so they have more time to study or play.

The findings of this research will contribute to the broader literature on child labor by providing evidence-based insights into how education-based interventions at the household level can help mitigate child labor. By highlighting the role of parental education in shaping children's work and schooling outcomes, this study offers valuable implications for governments, international organizations, and policymakers seeking to eliminate child labor and promote inclusive education. The results show a strong negative relationship between parent education and the number of hours a child works in a week.

Literature Review

The International Labor Organization refers to child labor as work that deprives children of their childhood, potential, and dignity while hindering their physical and mental development (ILO, 2024). Child labor remains a significant issue in many developing countries, influencing economic development through its impact on human capital, poverty, and long-term growth. Different countries might refer to child labor differently.

Parental education plays a critical role in child labor prevalence. Parents with lower education tend to involve their children in labor. A study in Côte d'Ivoire found that the absence of adequate school infrastructure forces parents to prioritize child labor over education (Abou, 2016). Thaddeus et al., 2022 study on gender disparities found that mothers with higher education tend to keep their daughters in school rather than send them to work. Akrono, 2025, using multi-stage sampling techniques in Ghana demonstrated that parental attitudes and education levels significantly influence whether children work, with better-educated parents less likely to push their children into labor.

In Ghana, economic contributions from child laborers are guised as apprenticeships but fall under exploitative labor conditions (Dunne & Humphreys, 2022). Evans and Skovdal, 2015 classify children's work based on whether it is paid or unpaid, with unpaid work frequently overlooked in policy discussions. The prevalence of unpaid domestic labor, often classified as informal child labor, remains high in Sub-Saharan Africa (Gatsinzi & Hilson, 2022).

In Sub-Saharan Africa, child labor is more prevalent in regions with widespread economic vulnerability and limited labor market opportunities (Krauss, 2016). A study of artisanal mining in Ghana illustrates how child labor often arises as a coping mechanism for families facing economic distress (Gatsinzi, 2019). Child labor spans multiple industries, with the majority engaged in agriculture, domestic work, and mining. Evans and Skovdal, 2015 defines child labor as any work that deprives children of schooling or exposes them to hazardous environments. In countries like Ghana and Burkina Faso, children commonly work in mining

and cocoa production, with many exposed to harmful chemicals (Odijie, 2020). The health effects ultimately outweigh the economic benefit of children's early entry into the labor market.

The consequences of child labor extend beyond economic inefficiency to educational and health setbacks. A study in Ethiopia found that children engaged in labor suffer from educational disruptions, leading to lower lifetime earnings (Nigussie, 2019). Another study highlights that child labor leads to poor health outcomes, as children are more prone to injuries and illnesses from hazardous work environments (Adedeji & Akinlo, 2016). Furthermore, long-term economic disadvantages emerge as child laborers have difficulty transitioning into higher-skilled occupations, reinforcing intergenerational poverty (Lambon-Quayefio & Owoo, 2018). Makworo, 2024 emphasized that child labor negatively affects school attendance and academic performance, often leading to long-term economic disadvantages. Basu, 2003 confirms that economic development reduces child labor by improving household incomes and increasing access to education. Conversely, child labor slows economic growth by limiting human capital accumulation. A review of past studies shows a research gap and justifies the need to analyze the influence of parental education attainment on the time children spend working.

Data & Methodology

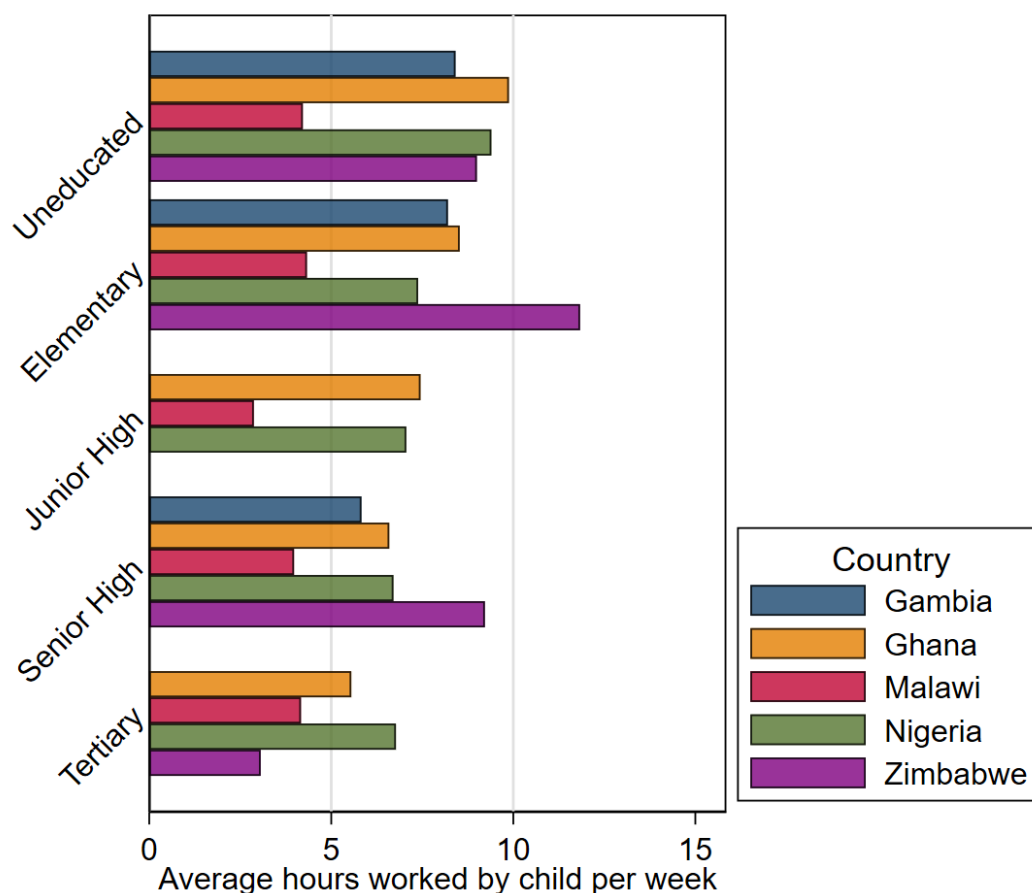
I use data from UNICEF's Multiple Indicator Cluster Surveys (MICS) version 6, released between 2019-2021 (Spencer et al., 2024). The MISC V6 expands on previous rounds by incorporating updated modules on early childhood development, education, family welfare, among other topics. The survey uses a two-stage stratified cluster sampling design to ensure national representativeness (UNICEF, 2021). Data on child labor, reported by the household respondent for children (0–17 years), were extracted for *Gambia, Ghana, Malawi, Nigeria and Zimbabwe*.

Table 1: Summary Statistics

	mean	min	max	sd
Age of child	10.38	5	17	3.72
Hours worked in past week	7.37	0	80	9.90
Number of children age 5-17	2.64	1	39	1.83
Number of children age < 5	1.03	0	20	1.12
Combined wealth score	-0.06	-2.96	5.33	0.97

Source: UNICEF Multi-Indicator Cluster Survey - Version 6.

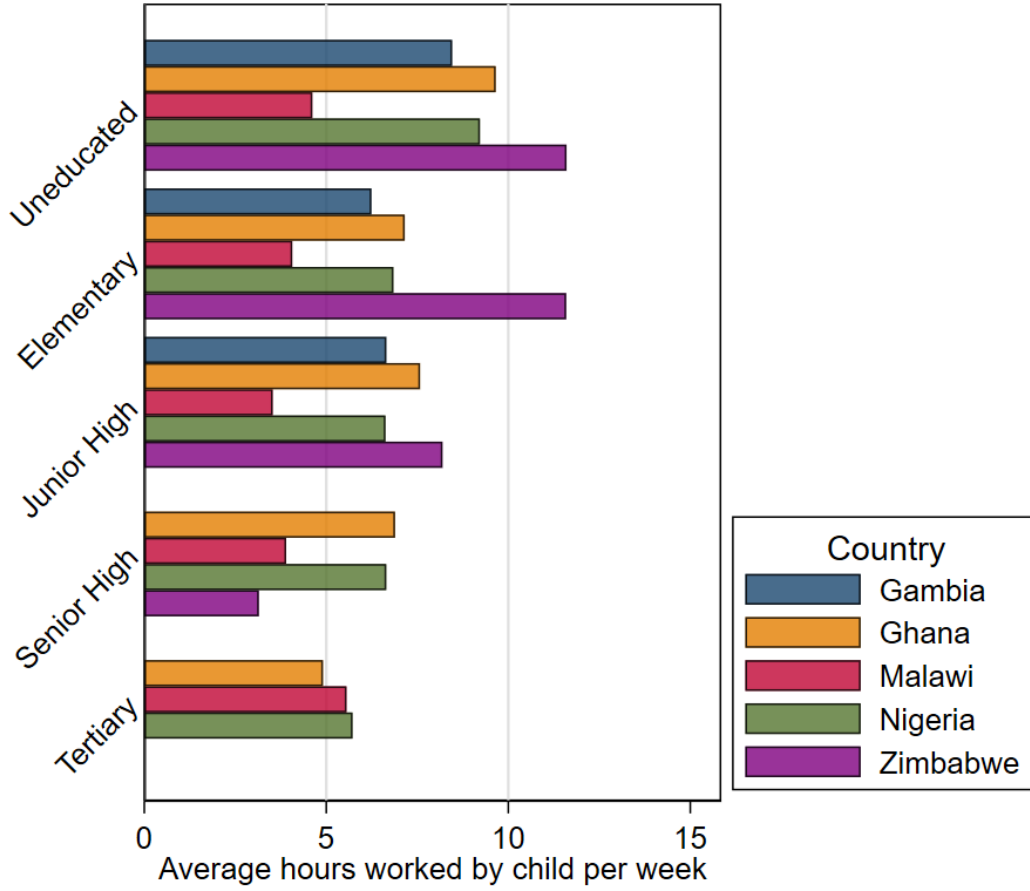
Figure 1: Average hours worked a week by father's education level



Source: UNICEF Multiple Indicator Cluster Survey (MICS) - Version 6

There are 29,602 children in the dataset with 2,541 from *Gambia*, 3,638 from *Ghana*, 6,824 from *Malawi*, 15,005 from *Nigeria* and 1,594 from *Zimbabwe*. The variable of interest is the number of hours a child works in a week. Figures 1 and 2 show that higher levels of education in parents equate to fewer hours worked per week by their child. The age of a child could influence the number of hours worked in a week due to physical development, so child age is included as a control. Figure 3 in the appendix shows that at 17 years, children work more hours in a week. Family wealth score, religion of the head of the household and child's gender are included as controls. Thaddeus et al., 2022 conducts a study on the female labor force and found that mothers with more advanced degrees are more likely to keep their daughters in school rather than send them to work. The data shows that on average children work up to 7 hours a week.

Figure 2: Average hours worked a week by mother's education level



Source: UNICEF Multiple Indicator Cluster Survey (MICS) - Version 6

Microeconomic Framework

Child's Utility Function

We begin by modeling a child's utility as a function of how they allocate their time between labor/chores and school/play activities:

$$U_c = f(L, S) \quad (1)$$

where U_c represents the child's utility, L is the time spent on labor or household chores, and S is the time spent on schooling and play activities. We assume that the child's utility increases with schooling and play ($\frac{\partial U_c}{\partial S} > 0$) and decreases with labor ($\frac{\partial U_c}{\partial L} < 0$).

The child faces a time constraint:

$$T = L + S \quad (2)$$

where T is the total time available.

Parental Control Model

However, children do not independently make decisions about their time allocation. Instead, parents control or heavily influence how children spend their time. We can model this by expressing the child's time allocation as functions determined by the parents:

$$L = g(E_p, I_p, H) \quad (3)$$

$$S = T - L = T - g(E_p, I_p, H) \quad (4)$$

where:

- E_p represents parental education level
- I_p represents parental income
- H represents household characteristics

Based on the empirical findings:

$$\frac{\partial L}{\partial E_p} < 0 \quad (5)$$

indicating that higher parental education levels are associated with fewer hours of child labor.

Extended Parental Decision Model

Parents make decisions about their child's time allocation by maximizing a utility function that includes both current household welfare and their perception of the child's welfare:

$$U_p = \alpha \cdot W_c + (1 - \alpha) \cdot W_h \quad (6)$$

where:

- U_p is the parent's utility
- W_c is the parent's perception of the child's welfare
- W_h is the household welfare (including income, consumption)
- α is the weight placed on the child's welfare relative to household welfare

The child's welfare as perceived by the parent can be expressed as:

$$W_c = \beta \cdot U_c + (1 - \beta) \cdot F_c \quad (7)$$

where U_c is the child's current utility, F_c is the parent's expectation of the child's future welfare, and β is the weight placed on current versus future welfare.

The parent's perception of how schooling affects future welfare can be modeled as:

$$F_c = h(S) \quad (8)$$

with $\frac{\partial F_c}{\partial S} > 0$, indicating that parents believe more schooling leads to better future outcomes.

Importantly, both α and β are likely functions of parental education:

$$\alpha = j_1(E_p) \quad \text{with} \quad \frac{\partial \alpha}{\partial E_p} > 0 \quad (9)$$

$$\beta = j_2(E_p) \quad \text{with} \quad \frac{\partial \beta}{\partial E_p} < 0 \quad (10)$$

suggesting that more educated parents place greater weight on their child's welfare relative to household welfare and are more future-oriented in their decision-making.

Econometric Specification

Based on this microeconomic framework, the econometric specification is:

$$L_i = \beta_0 + \beta_1 E_{p,i} + \beta_2 I_{p,i} + \beta_3 H_i + \varepsilon_i \quad (11)$$

L_i represents time spent working, $E_{p,i}$ parental education level, $I_{p,i}$ parental income, H_i household characteristics, i indexes children, and β_1 is expected to be negative based on the theory.

Specifically using the MISC6 data,

$$\begin{aligned} \log(CL_{hours}) = & \beta_0 + \beta_1 felevel + \beta_2 melevel + \delta_3 childage \\ & + \delta_4 headreligion + \delta_5 children5 \\ & + \delta_6 children17 + \delta_7 familywealth \end{aligned} \quad (12)$$

where:

CL_{hours} = weekly hours spent on labor

$felevel$ = father's level of education

melevel = mother's level of education

childage = child's age

children5 = No. of children below 5

children17 = No. of children below 17

headreligion = religion of household head

familywealth = wealth score for household

β_{1-2} = coefficients of key explanatory variables

δ_{3-7} = coefficients of control variables

Model Selection

The model is estimated using Ordinary Least square regression with fixed effects and a log-transformed dependent variable to adjust for the functional form. The log transformation provides intuitive percentage-based interpretations and also fixes abnormality in the data.

Findings

The microeconomic model (equation 6) posits that parents make decisions about their children's time allocation between labor (L) and schooling/play (S) based on a utility function that balances household welfare with the child's welfare. The empirical finding that $\frac{\partial L}{\partial E_p} < 0$ (equation 5) is strongly supported by the regression results, which show negative coefficients for all levels of parental education, with the father's tertiary education reducing child labor by approximately 11.9% and the mother's elementary education reducing it by approximately 8.4%. This aligns with equation 9 where $\alpha = j_1(E_p)$ with $\frac{\partial \alpha}{\partial E_p} > 0$, indicating that more educated parents place greater weight on their child's welfare relative to household consumption.

Similarly, equation 10 suggests $\beta = j_2(E_p)$ with $\frac{\partial \beta}{\partial E_p} < 0$, meaning educated parents are more future-oriented in their decision-making. This explains why parental education reduces child labor hours—educated parents perceive a stronger link between schooling (S) and future welfare (F_c) as modeled in equation 8: $F_c = h(S)$ with $\frac{\partial F_c}{\partial S} > 0$.

The positive coefficient for Muslim households (+17.0%) may reflect different values of α (weight placed on child welfare) or different perceptions of how schooling affects future welfare. The positive coefficients for child's age (+6.2% per year) and number of children (+2.1% per additional school-age child) align with household resource constraints from equation 2: $T = L + S$, where total available resources must be distributed among family members.

The negative wealth score coefficient (−9.1%) corresponds to I_p (parental income) in equation

Table 2: Regression Results

Dependent Variable: log(Child Weekly Hours Worked)	
Father's Education- Base:No Education	
Elementary	-0.008 (0.024)
Junior High	-0.030 (0.051)
Senior High	-0.056 (0.029)
Tertiary	-0.119* (0.048)
Mother's Education- Base:No Education	
Elementary	-0.084** (0.027)
Junior High	-0.067* (0.025)
Senior High	-0.056* (0.021)
Tertiary	-0.056 (0.071)
Religion - Base:Other Religion	
Muslims	0.170*** (0.027)
Child's age	0.062*** (0.003)
No. of children(age<5)	0.014* (0.006)
No. of children(age 5-17)	0.021*** (0.001)
Wealth Score	-0.091* (0.033)
Constant	0.810*** (0.051)
Fixed Effects	Yes
Observations	12794
R ²	0.073
Adjusted R ²	0.072
RMSE:0.935	
Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$	
Estimated using a Robust F.E Linear Regression	
Dependent variable is the natural logarithm of weekly hours worked	
Coefficients represent percentage changes.	

3: $L = g(E_p, I_p, H)$, demonstrating that as household income increases, the necessity for child labor decreases, allowing more time for schooling and play (S).

This econometric analysis empirically validates the microeconomic model of parental decision-making about child time allocation, confirming that parental education operates through both increased valuation of child welfare and stronger future orientation.

Conversely, this study uses only parental and household characteristics to evaluate child

labor, ultimately ignoring other important factors affecting child labor. This omitted variable bias poses significant challenges to the study by potentially distorting coefficient estimates and statistical inferences. When variables correlating with both parental education and child labor hours are excluded from the model, the estimated effects of parental education may be biased, with the direction depending on the specific correlations involved. This likely contributes to the model's relatively low R^2 (0.073), indicating substantial unexplained variation. For future analysis, incorporating regional fixed effects, measures of school quality, economic conditions, and cultural factors beyond religion would reduce the omitted variable bias.

Conclusion

This study demonstrates the critical role of parental education in reducing child labor across five African countries. Our regression analysis reveals that maternal education, particularly at the elementary level, serves as a significant protective factor—reducing weekly child labor hours by approximately 8.4%. Similarly, fathers with tertiary education have children working 11.9% fewer hours compared to uneducated fathers. These findings align with our microeconomic framework, which posits that educated parents place greater weight on their children's future welfare and recognize higher returns to education.

Beyond parental education, our results identify several other important determinants of child labor. Religious factors show significant influence, with Muslim households associated with 17% more child labor hours per week compared to Christian households. Demographic characteristics reveal that child labor increases with the child's age (6.2% more hours per additional year) and with more school-age children in the household (2.1% increase per additional child aged 5-17). Economic status emerges as a crucial factor, with each unit increase in household wealth score significantly reducing child labor hours by 9.1%.

Despite identifying these important relationships, our model's relatively low R^2 (0.073) suggests that while these factors matter, additional unobserved variables also contribute substantially to child labor practices. These findings point to potential intervention strategies focused on basic maternal education, targeted economic support for vulnerable households, and engagement with religious communities to address social norms around child labor. Future research should explore these additional factors and evaluate the effectiveness of multi-faceted interventions that address both educational and economic dimensions of child labor in sub-Saharan Africa.

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Appendix

Residual Analysis of Base Model

To provide evidence of statistical validity, we first analyze the residuals of the base model and check for violations of OLS. After careful review, it is evident that the base model violates several assumptions.

Base Model Residual Diagnostic

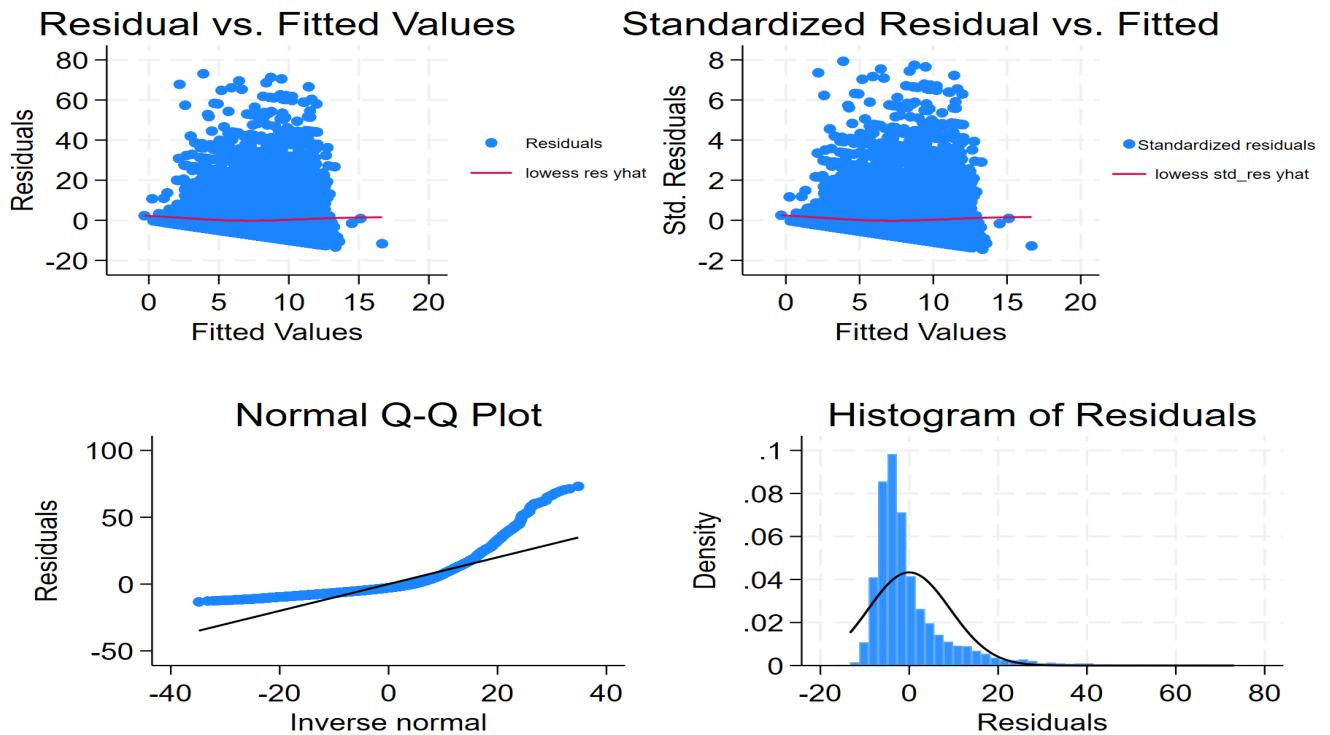


Table 3: Heteroskedasticity Test Results

Test	Test Statistic	df	p-value
Breusch-Pagan/Cook-Weisberg	$\chi^2(13) = 850.87$	13	< 0.0001
White's test	$\chi^2(83) = 217.98$	83	< 0.0001
Cameron & Trivedi's decomposition of IM-test			
Heteroskedasticity	$\chi^2 = 217.98$	83	< 0.0001
Skewness	$\chi^2 = 313.20$	13	< 0.0001
Kurtosis	$\chi^2 = 106.70$	1	< 0.0001
Total	$\chi^2 = 637.89$	97	< 0.0001

Note: Both the Breusch-Pagan and White tests strongly reject the null hypothesis of homoskedasticity (constant variance), indicating heteroskedasticity is present in the model. The Cameron & Trivedi decomposition further reveals issues with the normality of the error terms as shown by significant skewness and kurtosis. Robust standard errors should be used for inference, and model specification may need to be reconsidered.

Table 4: Model Specification Tests

Test	Test Statistic	p-value
Ramsey RESET test	$F(3, 12777) = 9.46$	0.0000
Linktest Results		
_hat	0.142 (0.186)	0.446
_hatsq	0.058 (0.012)	0.000
_cons	2.842 (0.664)	0.000

Note: Both tests indicate model misspecification. The significant Ramsey RESET test ($p < 0.001$) suggests omitted variables or incorrect functional form. The linktest confirms this with a significant _hatsq term ($p < 0.001$), indicating nonlinearity or missing higher-order terms. Model respecification is recommended, possibly including squared terms, interaction effects, or additional variables.

The base model shows severe issues, violating most of the OLS assumptions (linearity, heteroskedasticity, normality). This hinders interpretability, validity, and generalization of results.

Table 6 shows the steps taken;

Table 5: Linear Regression Assumptions and Remedial Actions

Violated Assumption	Fix
1. Non-linearity in the relationship between dependent and independent variables	Transformed dependent variable to the natural log. ($dv = \log(dv+1)$)
2. Non-random sample selection	Not violated as the sample used aligns with random sampling
3. Multicollinearity	V.I.F indicated no necessary adjustment is needed
4. Zero-Conditional Mean	Included interaction term; fixed effects
5. Heteroskedasticity	Used robust standard errors
6. Non-normality of error terms	Transformed dependent variable; robust regression; bootstrap standard errors

Residual Analysis of Final Model

Final Model Residual Diagnostic

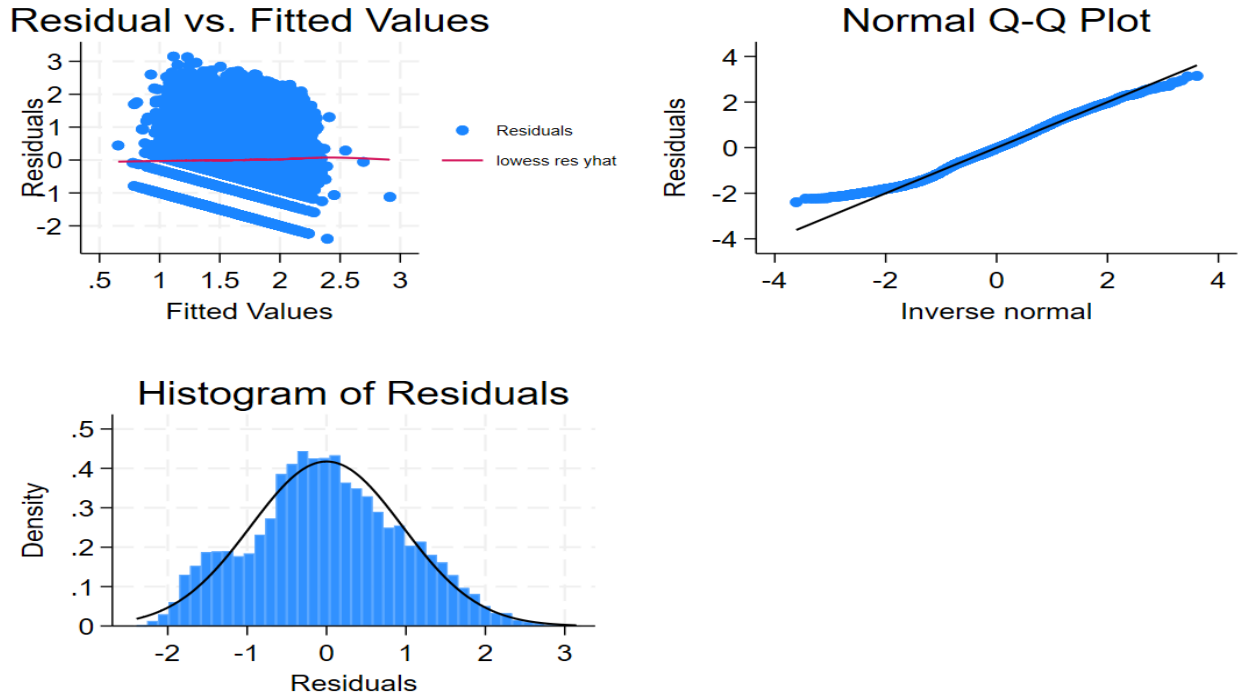


Figure 3: Final Model Diagnostics

The final model shows great improvement in terms of the OLS assumptions, indicating a good fit, and more importantly unbiased results appropriate for generalization.

Robustness And Sensitivity Checks

The best model is selected on the basis of the analysis of residuals, AIC, BIC, R^2 and in alignment with the assumptions of the OLS regression. This ensures the results interpreted in this paper are unbiased in any direction. The log FE Robust Std. Error model is selected because it corrects the functional form by taking the log of the dependent variable. This step is important because all previous models without the log dependent variable show a wrong function form in the residual versus fitted plot. These models are however perfect for a robustness checks. They all show strong conformity to the theoretical model of this paper where $\beta_i < 0$.

Table 6: Robustness Checks

Dependent Variable: Child Weekly Hours Worked					
	FE-Robust	FE-Bootstrap	GLM-Gamma	GLS-Het	FE-Log
Father's Education - Base:No Education					
Elementary	-0.386 (0.182)	-0.386 (0.288)	-0.032 (0.035)	-0.745*** (0.243)	-0.008 (0.024)
Junior High	-0.722 (0.341)	-0.722** (0.324)	-0.128** (0.052)	-0.938** (0.365)	-0.030 (0.051)
Senior High	-0.669** (0.205)	-0.669** (0.325)	-0.080* (0.042)	-0.181 (0.285)	-0.056 (0.029)
Tertiary	-1.119 (0.553)	-1.119 (1.233)	-0.125** (0.051)	-0.327 (0.371)	-0.119* (0.048)
Mother's Education - Base:No Education					
Elementary	-0.890** (0.227)	-0.890*** (0.321)	-0.125*** (0.034)	-1.274*** (0.233)	-0.084** (0.027)
Junior High	-0.753** (0.258)	-0.753*** (0.205)	-0.117** (0.050)	0.017 (0.337)	-0.067* (0.025)
Senior High	-0.287 (0.162)	-0.287 (0.180)	-0.051 (0.050)	0.015 (0.332)	-0.056* (0.021)
Tertiary	-0.525 (0.571)	-0.525 (1.363)	-0.114 (0.079)	-0.222 (0.528)	-0.056 (0.071)
Religion - Base:Christian					
Muslims	1.799*** (0.216)	1.799*** (0.609)	0.232*** (0.031)	2.202*** (0.203)	0.170*** (0.027)
Age of child	0.452*** (0.051)	0.452*** (0.068)	0.062*** (0.004)	0.417*** (0.024)	0.062*** (0.003)
Number of children under age 5	0.081 (0.088)	0.081 (0.110)	0.011 (0.010)	0.076 (0.079)	0.014* (0.006)
Number of children age 5-17	0.164*** (0.015)	0.164*** (0.034)	0.020*** (0.006)	0.160*** (0.043)	0.021*** (0.001)
Combined wealth score	-0.564 (0.281)	-0.564* (0.334)	-0.068*** (0.020)	-0.835*** (0.110)	-0.091* (0.033)
Constant	1.381 (0.849)	1.381*** (0.356)	1.009*** (0.081)	1.242*** (0.395)	0.810*** (0.051)
Observations	12794	12794	12794	12794	12794
R-squared	0.049	0.049			0.073
AIC	92924.258	92944.258	75050.045	.	34589.491
BIC	92954.085	93048.652	75184.266	.	34619.318

Standard errors in parentheses

FE-Robust = Fixed Effects with Robust Standard Errors

FE-Bootstrap = Fixed Effects with Bootstrapped Standard Errors

GLM-Gamma = Generalized Linear Model with Gamma Distribution

GLS-Het = Generalized Least Squares with Heteroskedastic Panels

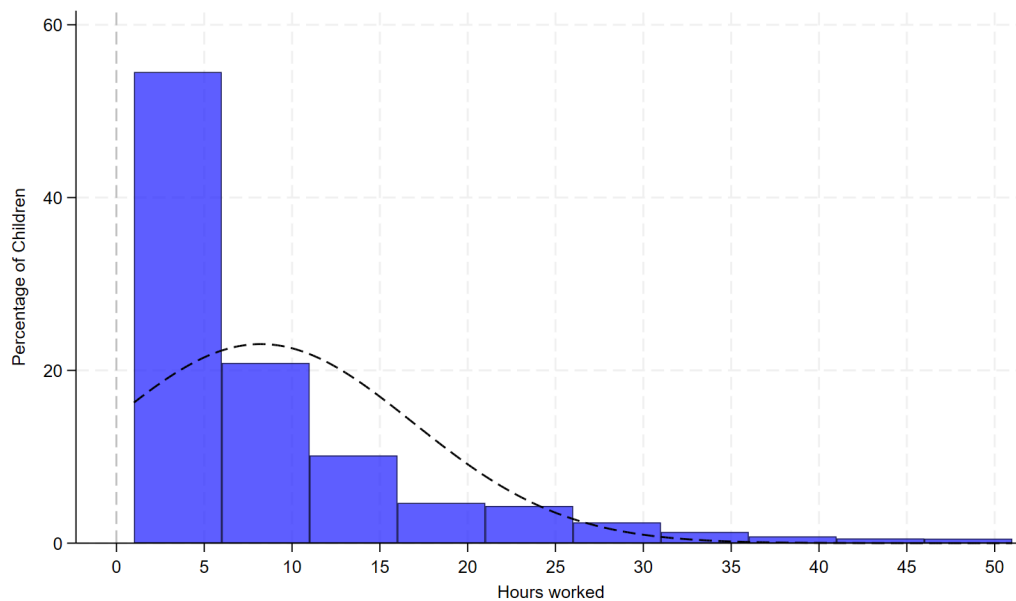
FE-Log = Fixed Effects on Log-Transformed Dependent Variable

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7: Summary of Econometric Models

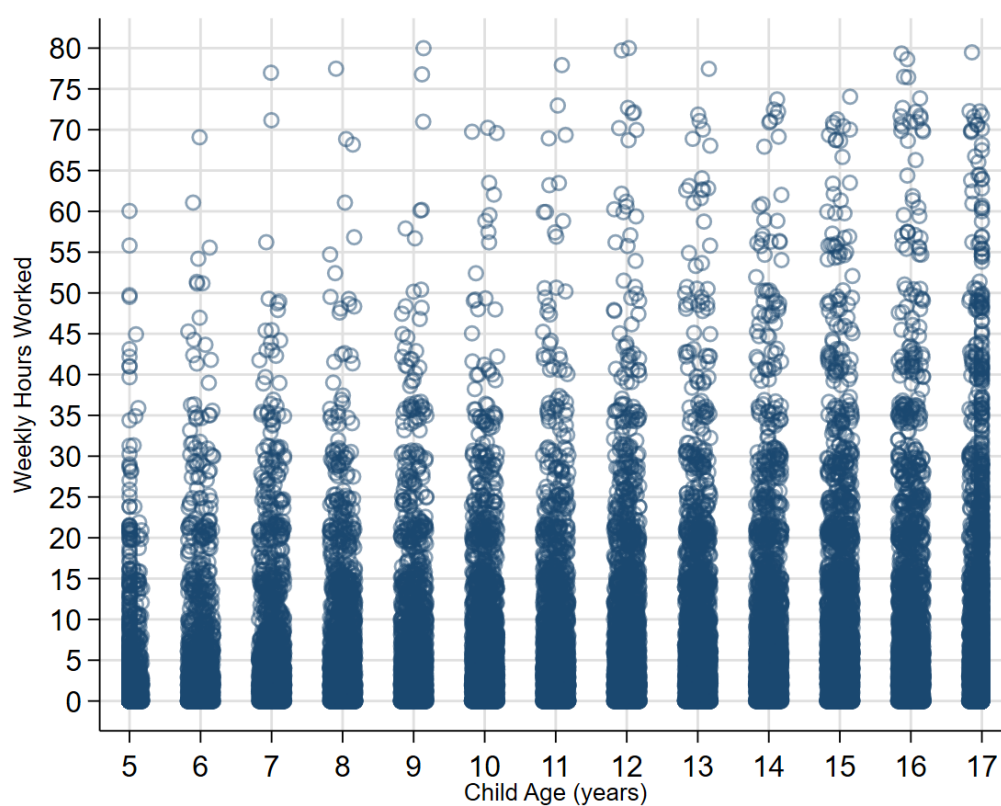
Model	Purpose and Characteristics
Base OLS	Standard regression approach serving as benchmark
FE-Robust	Fixed Effects model with robust standard errors to account for heteroskedasticity
FE-Bootstrap	Fixed Effects with bootstrapped standard errors to address unobserved heterogeneity, heteroskedasticity, and non-linearity
GLM-Gamma	Generalized Linear Model with Gamma distribution to accommodate non-normality and skewness in the dependent variable
GLS-Het	Generalized Least Squares to account for heteroskedasticity and serial correlation in panel data
FE-Log	Fixed Effects with log-transformed dependent variable to stabilize variance and model proportional relationships
Note: Multiple model specifications were used to validate results under different statistical assumptions and robustness check	

Figure 4: Distribution of Number of hours worked



Source: UNICEF Multi-Indicator Cluster Survey - Version 6.

Figure 5: Number of Hours worked at different child ages



Source: UNICEF Multi-Indicator Cluster Survey - Version 6.